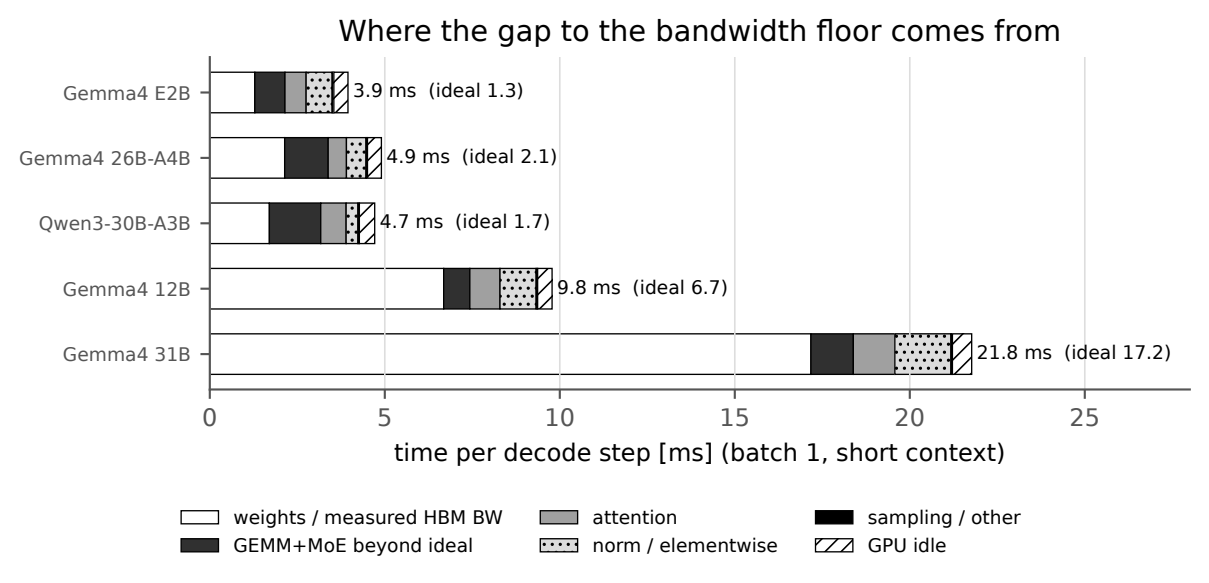

Where the gap to the bandwidth floor comes from
Gemma4 E2B
3.9 ms (ideal 1.3)
Gemma4 26B-A4B
4.9 ms (ideal 2.1)
Qwen3-30B-A3B
4.7 ms (ideal 1.7)
Gemma4 12B
9.8 ms (ideal 6.7)
Gemma4 31B
21.8 ms (ideal 17.2)
0
5
10
15
20
25
time per decode step [ms] (batch 1, short context)
weights / measured HBM BW
GEMM+MoE beyond ideal
attention
norm / elementwise
sampling / other
GPU idle